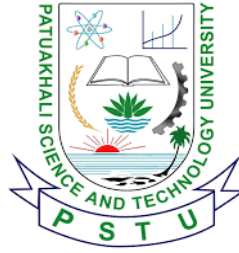


PATUAKHALI SCIENCE AND TECHNOLOGY UNIVERSITY



Assignment - 1

Course Title : Network Routing & Switching
Course Code : CCE-415

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Level – 4 , Semester - 1

Faculty of Computer Science and
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1.Primary Functions and Features of a Router.

Functions:

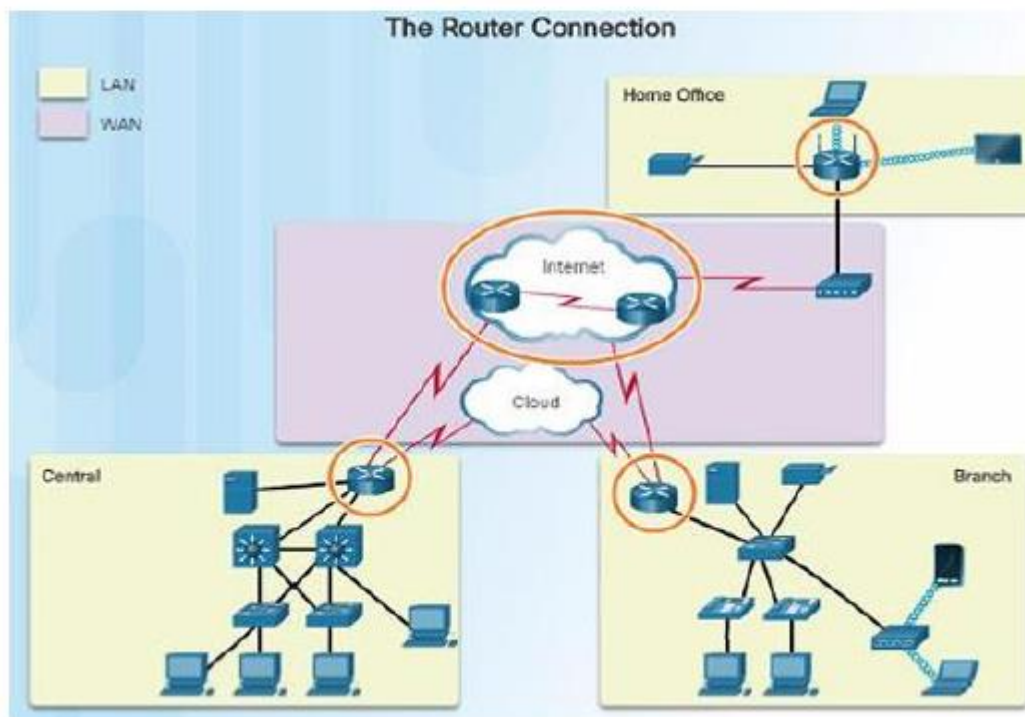
- Connects multiple networks (LANs, WANs).
- Determines the best path for data packets.
- Routes traffic based on destination IP addresses (Layer 3 device).
- Provides security features like ACLs (Access Control Lists), NAT (Network Address Translation), and firewall rules.

Features:

- Multiple interfaces for connecting different networks.
- Support for static and dynamic routing.
- Packet switching and filtering.
- QoS (Quality of Service) for traffic prioritization.
- Support for VPNs and remote access.

2. Connect Devices for a Small Routed Network.

- ✓ Router is responsible for forwarding packets from network to network, from the source to the destination. A router connects multiple networks, which means that it has multiple interfaces that each belong to a different IP network. These interfaces are used to connect
 - ✧ LANs – Ethernet networks that contain PCs, printers, and servers
 - ✧ WANs – used to connect networks over large geographical areas such as to an ISP
- ✓ When a packet arrives on a router's interface, the router might be the final destination, or it may have to send it to another router to reach its final destination.



3. Configure basic settings on a router to route between two directly-connected networks, using CLI.

Configure Basic Router Settings (1.1.3.1)

Cisco routers and Cisco switches are a lot alike. They support a similar modal operating system, similar command structures, and many of the same commands. In addition, both devices have similar initial configuration steps.

For instance, the following configuration tasks should always be performed:

- **Name the device**—Distinguishes it from other routers.
- **Secure management access**—Secures privileged EXEC, user EXEC, and remote access.
- **Configure a banner**—Provides legal notification of unauthorized access.

Always save the changes on a router and verify the basic configuration and router operations.

Figure 1-20 shows the topology used for example configurations.

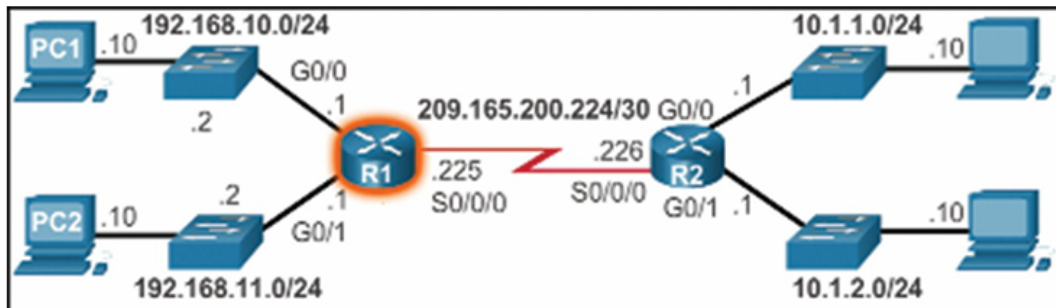


Figure 1-20 IPv4 Configuration Topology

```
Router# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)# hostname R1
R1(config)# enable secret class
R1(config)# line console 0
R1(config-line)# password cisco
R1(config-line)# login
R1(config-line)# exit
R1(config)# line vty 0 4
R1(config-line)# password cisco
R1(config-line)# login
R1(config-line)# exit
R1(config)# service password-encryption
R1(config)# banner motd $ Authorized Access Only! $
R1(config)# end
R1# copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R1#
```

4. Verify connectivity between two networks that are directly connected to a router.

1. show ip interface brief

Check Interface Configuration. This will show Interface status (up/down)

2. show ip route

Verify Routing Table. Check if the router has routes for both directly connected networks.

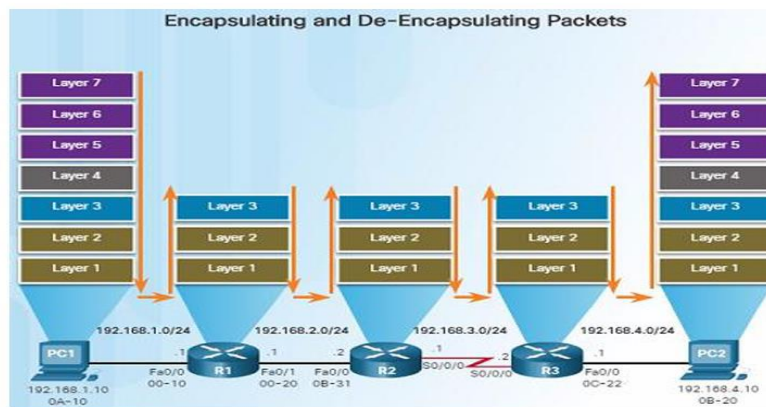
3. ping 192.168.2.10

Test Connectivity Using Ping: From a device in **one network**, ping a device in the **other network**.

4. traceroute 192.168.2.10

Use Traceroute for Path Testing

5. Explain the encapsulation and de-encapsulation process used by routers when switching packets between interfaces.



PC1 is configured with IPv4 address 192.168.1.10 and an example MAC address of 0A-10. As a packet travels from the source device to the final destination device, the Layer 3 IP addresses do not change. However, the Layer 2 data link addresses change at every hop as the packet is de-encapsulated and re-encapsulated in a new Layer 2 frame by each router.

This diagram and explanation illustrate how data travels across a network from one device (PC1) to another (PC2), passing through multiple routers (R1, R2, R3). Here's a breakdown of what's happening:

Layer 3 (IP Address)

- The **Layer 3 IP address** is the logical address used to identify devices on different networks.
- In this example, **PC1** has the IP address **192.168.1.10** and **PC2** has **192.168.4.10**.
- As the packet travels from PC1 to PC2, **the source and destination Layer 3 IP addresses remain the same** throughout the journey, regardless of how many routers (hops) the packet passes through.

Layer 2 (MAC Address)

- The **Layer 2 address (MAC address)** is used for communication within the same local network segment.
- As the packet moves from one network segment to another (from one router to the next), **the Layer 2 addresses change at each hop**.
- Each router de-encapsulates the incoming frame (removes the old Layer 2 header), examines the Layer 3 IP address to determine where to send the packet next, and then re-encapsulates the packet with a new Layer 2 header appropriate for the next segment.

When a router receives a packet from one network that is destined for another network, the router performs the following three steps:

Step	Process	Purpose
Step 1	De-encapsulation of Layer 2 frame	Exposes the Layer 3 IP packet.
Step 2	Examines destination IP & checks routing table	Determines the best next-hop or exit interface.
Step 3	Re-encapsulation in a new Layer 2 frame & forwarding	Sends the packet to the next destination.

6. Explain the path determination function of a router.

A primary function of a router is to determine the best path to use to send packets. To determine the best path, the router searches its routing table for a network address that matches the destination IP address of the packet.

The routing table search results in one of three path determinations:

- **Directly connected network**—If the destination IP address of the packet belongs to a device on a network that is directly connected to one of the interfaces of the router, that packet is forwarded directly to the destination device. This means that the destination IP address of the packet is a host address on the same network as the interface of the
- **Remote network**—If the destination IP address of the packet belongs to a remote network, then the packet is forwarded to another router. Remote networks can only be reached by forwarding packets to another router.
- **No route determined**—If the destination IP address of the packet does not belong to either a connected or a remote network, the router determines if there is a Gateway of Last Resort available. A Gateway of Last Resort is set when a default route is configured or learned on a router. If there is a default route, the packet is forwarded to the Gateway of Last Resort. If the router does not have a default route, then the packet is discarded.

The logic flowchart in [Figure 1-30](#) illustrates the router packet-forwarding decision process.

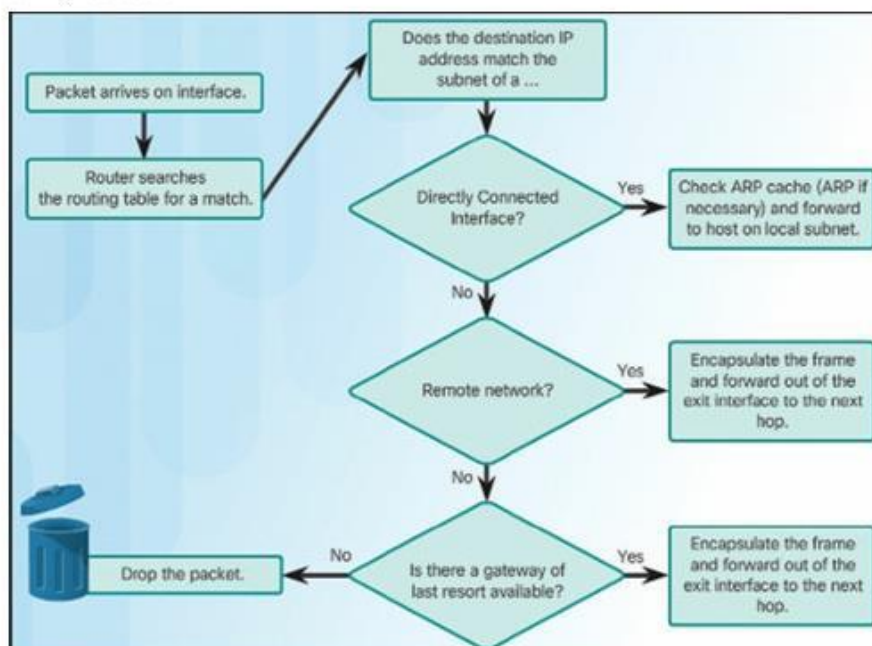


Figure 1-30 Packet-Forwarding Decision Process

7. Explain routing table entries for directly connected networks.

Directly Connected Routing Table Entries (1.3.2.2)

An active, properly configured, directly connected interface actually creates two routing table entries. [Figure 1-35](#) displays the IPv4 routing table entries on R1 for the directly connected network 192.168.10.0.

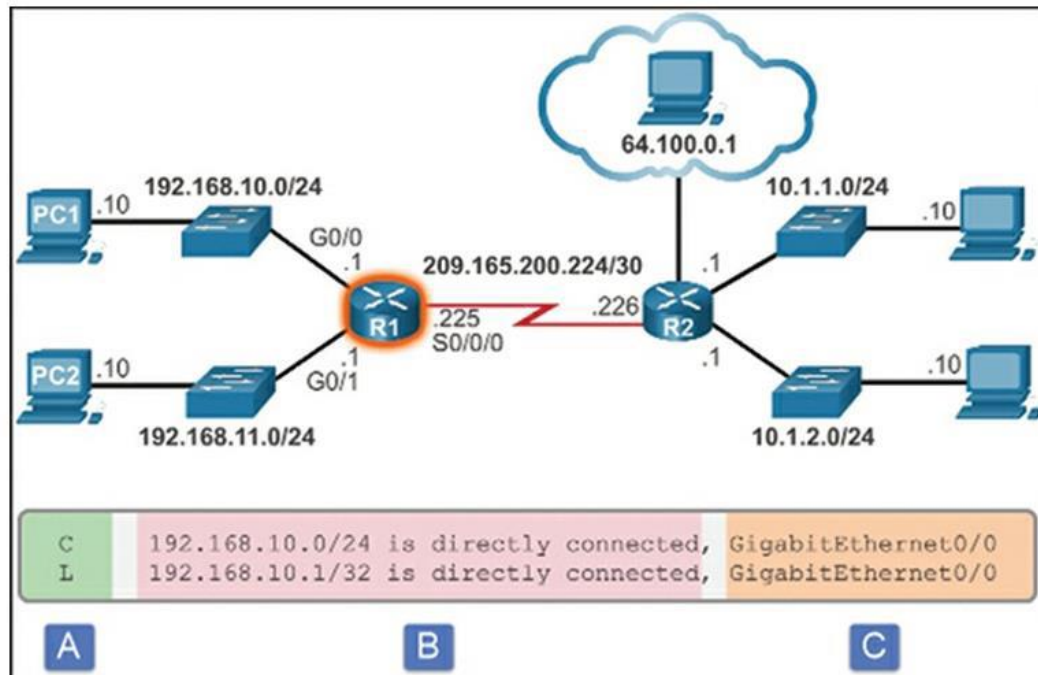


Figure 1-35 Directly Connected Network Entry Identifiers

Legend	Name	Description
A	Route Source	Identifies how the network was learned by the router. Directly connected interfaces have two route source codes. 'C' identifies a directly connected network. 'L' identifies the IPv4 address assigned to the router's interface.
B	Destination Network	Identifies the destination network and how it is connected.
C	Outgoing Interface	Identifies the exit interface to use when forwarding packets to the destination network.

8. Explain how a router builds a routing table of directly connected networks

A newly deployed router, without configured interfaces, has an empty routing table,

Before the interface state is considered up/up and added to the IPv4 routing table, the interface must

- Be assigned a valid IPv4 or IPv6 address
- Be activated with the **no shutdown** command
- Receive a carrier signal from another device (router, switch, host, and so on)

When the interface is up, the network of that interface is added to the routing table as a directly connected network.

9. Explain how a router builds a routing table using static routes.

Static routes are manually configured. They define an explicit path between two networking devices. Unlike a dynamic routing protocol, static routes are not automatically updated and must be manually reconfigured if the network topology changes. The benefits of using static routes include improved security and resource efficiency. Static routes use less bandwidth than dynamic routing protocols, and no CPU cycles are used to calculate and communicate routes. The main disadvantage to using static routes is the lack of automatic reconfiguration if the network topology changes.

There are two common types of static routes in the routing table:

- Static route to a specific network
- Default static route

A static route can be configured to reach a specific remote network. IPv4 static routes are configured using the following command:

[Click here to view code image](#)

```
Router(config)# ip route network mask { next-hop-ip |  
exit-intf }
```

A static route is identified in the routing table with the code 'S.'

10. Explain how a router builds a routing table using a dynamic routing protocol.

Routers share routing information using protocols (RIP, OSPF, EIGRP, BGP).

Example with OSPF:

```
router ospf 1  
network 192.168.1.0 0.0.0.255 area 0  
network 192.168.2.0 0.0.0.255 area 0
```

The router dynamically learns about other networks and adds them to its routing table:

```
O 192.168.3.0/24 [110/2] via 192.168.2.2
```